Case Report

FDG PET Scan in Cutaneous Rosai–Dorfman–Destombes Disease

Shrikant Vasantrao Solav, Chakor Vora, Rajlaxmi Jagtap, Shailendra Savale, Ishtiyaq Shaikh, Aman Solav

Spect Lab Nuclear Medicine Services, Pune, Maharashtra, India

Abstract

Rosai–Dorfman–Destombes (RDD) disease is also called as sinus histiocytosis and is characterized by enlarged lymph nodes and previously called as non-Langerhans cell histiocytosis. Based on pathologic, molecular, and genetic features, RDD disease has been classified into sporadic noncutaneous (classical nodal, extranodal, neoplasia associated, and autoimmune associated), familial (H syndrome, autoimmune lymphoproliferative syndrome related, and familial NOS), and cutaneous subtypes. Cutaneous RDD disease is not associated with lymphadenopathy or visceral organ involvement. The disease is usually localized and has relatively better long-term prognosis. Presented here is a case of indurated plaque-like skin lesions over the abdomen. ^{18}F-fluorodeoxyglucose (FDG) positron emission tomography-computed tomography scan revealed FDG avid cutaneous–subcutaneous soft-tissue lesions. Histology confirmed the diagnosis of cutaneous RDD disease.

Keywords: *Cutaneous Rosai–Dorfman–Destombes disease, fluorodeoxyglucose positron emission tomography scan, Rosai–Dorfman–Destombes*

Introduction

Rosai–Dorfman–Destombes (RDD) disease is a non-Langerhans cell histiocytosis characterized by the accumulation of activated histiocytes in various tissues ranging from lymph nodes, liver, spleen, central nervous system, mucosal surface of gastrointestinal tract, kidneys, and rarely bone marrow.[1]

Classical R-type RDD disease presents with nodal enlargement with or without systemic symptoms such as fever and weight loss. Usually, the cervical–axillary–mediastinal nodes are involved. Retroperitoneal nodes are less commonly involved.[2] Cutaneous involvement is seen in 10% of cases. Multisystem involvement is seen in 20% and prognosis is proportional to the number of sites involved. Isolated cutaneous involvement is rare[3] presented here is a case of isolated cutaneous RDD disease.

Case Report

A 43-year-old female developed thick plaque-like hyperpigmented skin-based lesions over the abdomen-thoraco-abdominal wall. Not associated with itching. There was no fever or loss of weight. Examination showed irregular dermal-based plaque-like skin thickening with nodularity. Skin biopsy showed features of RDD disease. ^{18}F-fluorodeoxyglucose positron emission tomography-computed tomography (^{18}F-FDG PET CT) scan was performed using 7 mCi of fluorine-18-FDG administered intravenously in 6 h fasting state. Images were acquired on united PET CT system. Images were reconstructed in various planes. The study revealed FDG avid cutaneous-based thickening over the right lateral thoracoabdominal wall. Smaller skin lesion was seen in the right high gluteal region. The right axillary nodes and left level II neck node-right level Va neck nodes were identified [Figure 1].

Histology of the lesion shows dense and diffuse infiltrates of lymphocytes and plasma cells, large foamy histiocytes are also seen spanning superficial and deep reticular dermis with loss of dermal appendage. The foamy histiocytes show emperipolesis. Some plasma cells are giant and multinucleated. The rest of the dermis shows dense fibroplasia. The epidermis shows mild hyperplasia and spongiosis. On immunohistochemistry, histiocytes express S100, CD68, and negative for CD1a consistent with RDD disease.

Discussion

RDD disease was initially described by Destombes in 1965 and later by Rosai and

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

How to cite this article: Solav SV, Vora C, Jagtap R, Savale S, Shaikh I, Solav A. FDG PET scan in cutaneous Rosai–Dorfman–Destombes disease. Indian J Nucl Med 2024;39:396-8.

Address for correspondence:
Dr. Shrikant Vasantrao Solav, Spect Lab Nuclear Medicine Services, Pune, Maharashtra, India.
E-mail: drsolav1@gmail.com

Received: 09-09-2024
Revised: 20-09-2024
Accepted: 07-10-2024
Published: 25-01-2025

Access this article online

Website: https://journals.lww.com/ijnm

DOI: 10.4103/ijnm.ijnm_121_24

Quick Response Code:

© 2025 Indian Journal of Nuclear Medicine | Published by Wolters Kluwer - Medknow

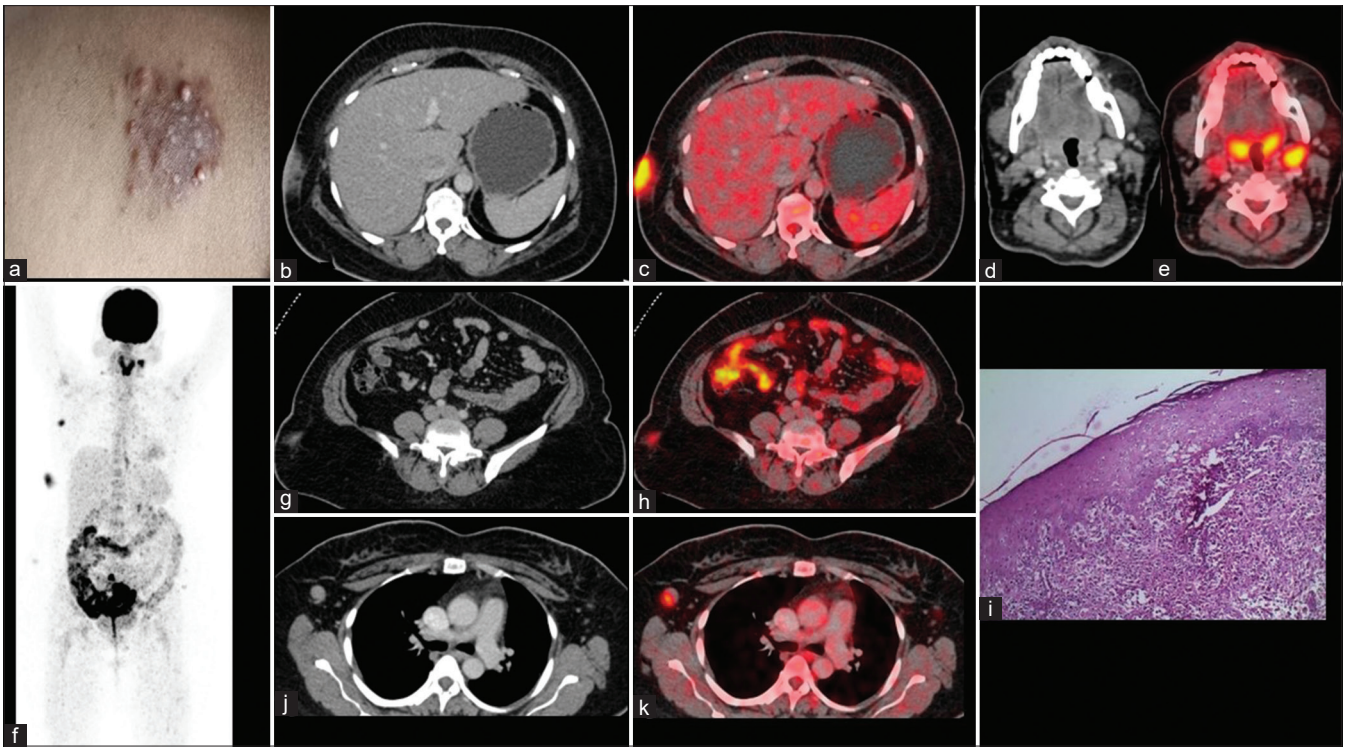


Figure 1: Clinical picture shows nodular - Hyperpigmented lesion over the lateral thoracoabdominal wall with irregular margins (a). Clinically considered to be either Hansen's disease or Actinomycosis. Skin biopsy showed dense and diffuse infiltrates of lymphocytes and numerous plasma cells and large foamy histiocytes. Infiltrate was spanning superficial and deep layers of the dermis. Foamy histiocytes in the center show emperipolesis (i). ^{18}F-fluorodeoxyglucose (FDG) positron emission tomography computed tomography showed FDG avid three foci of radiotracer uptake along the right side of the body (f). There corresponded to the cutaneous thickening seen clinically (b and c), An additional subcutaneous lesion was seen in the right high gluteal region (g and h). FDG avid right axillary (j and k) and left level II neck nodes were also noted (d and e)

Dorfman in 1969 as sinus histiocytosis with enlarged nodes.[4] The working group of histiocytosis initially classified this as non-Langerhans cell histiocytosis (1987) and recently reclassified it based on pathological-molecular-genetic features.[5] The revised classification divides this entity into (1) sporadic, (2) familial, and (3) cutaneous. The sporadic-noncutaneous could be subdivided into (a) isolated nodal disease, (b) extranodal (single abdominal viscera, skeletal or disseminated), (c) neoplasia associated (lymphoma, leukemia, malignant histiocytosis, Erdheim–Chester disease, and Langerhans cell histiocytosis). (d) Autoimmune disease associated (systemic lupus erythematosus, inflammatory juvenile arthritis, HIV, etc.) familial subtype could be H-syndrome, autoimmune lymphoproliferative, and familial Not otherwise specified (NOS). Cutaneous (C subtype) could be xanthogranuloma family (S100 negative) and nonanthogranuloma family.

Clinically patients present with fever, weight loss, night sweats, and enlarged nodes. Extranodal disease includes the following sites, skin (10%), bone marrow (5%–10%), orbit (11%), and nasal mucosa (11%).[6]

Cutaneous lesions of RDD disease present as nodular-popular rash/eruptions or plaques usually localized to the head-neck or extremities. Trunk involvement is rare. The lesions may or may not be tender.[7] Differential diagnosis of similar cutaneous lesions includes tuberculosis, Kaposi sarcoma, eruptive xanthoma, acneiform eruptions, cutaneous lymphoma, and Langerhans cell histiocytosis.[8]

^{18}F-FDG PET CT is helpful in localizing nodal as well as visceral organ involvement in RDD disease. The test may not be able to differentiate between lymphoma or similar pathologies and RDD disease.[9] Our patient had FDG avid cutaneous and subcutaneous thickening at two sites. There were a few lymph nodes in neck as well as the right axilla. Histology revealed phagocytosis of inflammatory cells in histiocytes which is called as emperipolesis (derived from Greek en-inside and peripoliomai-going around).

Acknowledgment

The authors would like to thank the team of technologists involved in acquiring the images: Ranjit Mahajan, Parag Deshmukh, Prathmesh, Bhagyashree, Ramdas, Shivam for their sincerity.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The

patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Foucar E, Rosai J, Dorfman R. Sinus histiocytosis with massive lymphadenopathy (Rosai-Dorfman disease): Review of the entity. Semin Diagn Pathol 1990;7:19-73.
2. Sodhi KS, Suri S, Nijhawan R, Kang M, Gautam V. Rosai-Dorfman disease: Unusual cause of diffuse and massive retroperitoneal lymphadenopathy. Br J Radiol 2005;78:845-7.
3. Al-Khateeb TH. Cutaneous Rosai-Dorfman disease of the face: A comprehensive literature review and case report. J Oral Maxillofac Surg 2016;74:528-40.
4. Rosai J, Dorfman RF. Sinus histiocytosis with massive lymphadenopathy. A newly recognized benign clinicopathological entity. Arch Pathol 1969;87:63-70.
5. Emile JF, Abla O, Fraitag S, Horne A, Haroche J, Donadieu J, *et al.* Revised classification of histiocytoses and neoplasms of the macrophage-dendritic cell lineages. Blood 2016;127:2672-81.
6. Brenn T, Calonje E, Granter SR, Leonard N, Grayson W, Fletcher CD, *et al.* Cutaneous Rosai-Dorfman disease is a distinct clinical entity. Am J Dermatopathol 2002;24:385-91.
7. Gaul M, Chang T. Cutaneous Rosai-Dorfman disease. Cutis 2019;103:171-3.
8. El-Kamel MF, Selim MK, Gawad MM. A new presentation of isolated cutaneous Rosai-Dorfman disease: Eruptive xanthoma-like lesions. Indian J Dermatol Venereol Leprol 2020;86:158-61.
9. Liu B, Lee NJ, Otero HJ, Servaes S, Zhuang H. Rosai-Dorfman disease mimics lymphoma on FDG PET/CT in a pediatric patient. Clin Nucl Med 2014;39:206-8.